

A) Choose an example model to explain the figure on slide 17 (1 pt) :

Breast Cancer Prediction using Naive Bayes

1. General Accuracy:

The Naive Bayes model should achieve high overall accuracy when predicting breast cancer risk using factors like age, obesity, physical activity, and medical history.

2. Accuracy for Subgroups:

The model should perform fairly across different groups such as married women, unmarried women, women before pregnancy, and women after pregnancy.

3. Predictive Power:

It should be able to accurately predict future unseen patient cases, not just the data used for training.

4. Applicability:

With enough diverse medical data, the model can be useful in real-world healthcare settings for early risk assessment.

5. Limitations:

The model may face challenges such as biased datasets, missing health information, or zero-probability issues if training data is insufficient.

B) Select an example to explain the visual analytics workflow on slide 22 (1 pt)

Problem:

Breast cancer deaths are increasing because cancer is often identified too late.

1. Define the subject of analysis:

The goal is to predict breast cancer risk early using patient health data.

2. Collect data:

Gather previous diagnosis records for patients, including attributes such as age, blood pressure, blood reports, sugar level, obesity, and other medical indicators.

3. Understand and preprocess the data:

Analyze data quality, identify missing values, remove noise, detect bias, and ensure the dataset is reliable.

4. Explore the data visually:

Use visual analytics techniques to study distributions, detect outliers, calculate averages or medians, and identify correlations between attributes.

5. Build analytical models:

Select suitable models such as Decision Trees, Logistic Regression, or SVM based on the dataset.

6. Evaluate the model:

Measure performance using accuracy, precision, recall, and error rates

7. Refine if needed:

If model performance is poor, revisit data quality, preprocessing steps, or model choice and improve the workflow.

8. Real-world application:

Deploy the validated model to support early breast cancer detection and improve diagnosis speed.

C) Explain the aspects that lead to enhanced visualization on slide 53 - 57 (1 pt)

- **Reproducibility:** Visual analytics should produce consistent results so others can repeat the same analysis, such as using the same patient dataset to reach similar medical predictions.
- **Guidance:** The system should support users with suggestions or directions, like recommending important sales trends during dashboard analysis.
- **Progressive Visual Analytics:** Analysis should improve step by step, for example starting with overall cancer rates and then drilling into age-specific patterns.
- **Pair and Cooperative Visual Analytics:** Collaboration between humans or human-AI systems improves insights, such as doctors and analysts working together on diagnosis data.
- **Scalability:** Visual analytics tools should handle large and complex datasets effectively, like analyzing millions of customer transactions in Power BI.

D) Explain the differences between quantitative, ordinal, and nominal data (1 pt) :

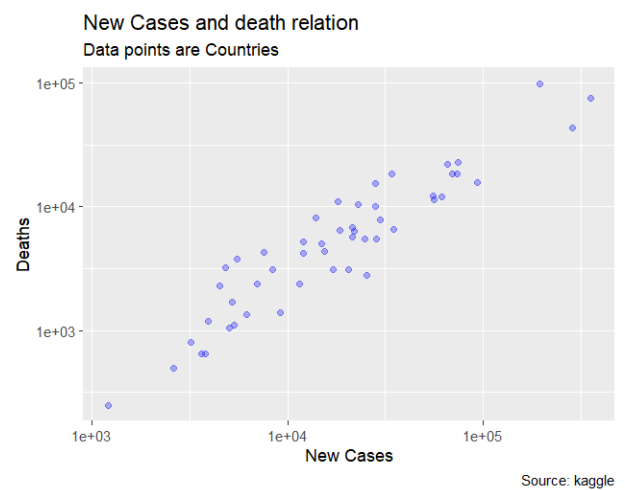
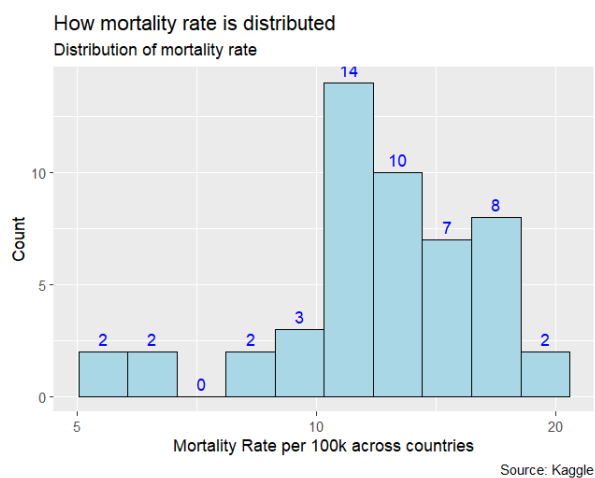
Quantitative data refers to numerical values that can be measured or counted, such as age, salary, height, or sales revenue.

Ordinal data is categorical data with a meaningful order or ranking, such as satisfaction levels (high, medium, low), not precise.

Nominal data is categorical data without any natural order, where categories are simply labels, such as colors (red, blue, green) or gender.

E) Given the four visual representations for distributions, co-distributions, spatial data, and temporal data on slide 79: Choose two representations, find different data examples for them, and create appropriate visualizations. Explain your data selection and visualization choices. You may use any dataset and tool. (3 pts)

Country	Region	Continent	Population_M	New_Cases_2022	Deaths_2022	Incidence_Rate_Per_100K	Mortality_Rate_Per_100K
United States	North America	Americas	331.9	287850	43250	89.4	13.0
China	East Asia	Asia	1412.0	357000	75000	29.0	5.4
India	South Asia	Asia	1408.0	195000	99000	25.8	12.7
Brazil	South America	Americas	214.3	73610	18360	46.8	11.9
United Kingdom	Northern Europe	Europe	67.3	55920	11400	93.6	14.2
Germany	Western Europe	Europe	83.2	69900	18570	90.5	16.5
France	Western Europe	Europe	67.8	61210	12150	99.1	15.6
Japan	East Asia	Asia	125.7	94000	15800	76.3	10.6
Australia	Oceania	Oceania	25.7	20600	3100	88.7	11.0
Canada	North America	Americas	38.2	28600	5500	83.5	12.2
Italy	Southern Europe	Europe	59.6	55700	12300	92.6	15.8
Spain	Southern Europe	Europe	47.4	34750	6600	78.9	11.6
South Korea	East Asia	Asia	51.7	25500	2800	64.2	6.4



1. Distribution – Histogram of Mortality Rate

- I used the breast cancer mortality rate per 100K to understand how mortality is distributed across countries.
- A histogram was the right choice because it clearly shows the spread, frequency, and variation of a single numerical variable.
- This helps identify common mortality ranges and possible outliers globally.

2. Co-distribution – Scatter Plot of New Cases vs Deaths

- I used new breast cancer cases and deaths to study their relationship across countries.
- A scatter plot was suitable because it shows how two numerical variables are related.
- This helps reveal that countries with more cases often tend to have higher death counts.

F) Using a dataset of your choice: Complete all steps described for understanding individual attributes (slides 21 - 27). Document and visualize your findings for each step. (3 pts)

Understanding Data:

STEP 1 — Look at what data is available

Name: Breast cancer by country dataset

The dataset contains **13 attributes**.

We will analyze: **Mortality_Rate_Per_100K**

```
'data.frame': 50 obs. of 13 variables:
 $ Country      : chr "United States" "China" "India" "Brazil" ...
 $ Region       : chr "North America" "East Asia" "South Asia" "South America" ...
 $ Continent    : chr "Americas" "Asia" "Asia" "Americas" ...
 $ Population_M : num 331.9 1412 1408 214.3 67.3 ...
 $ New_Cases_2022 : int 287850 357000 195000 73610 55920 69900 61210 94000 20600 28600 ..
 $ Deaths_2022  : int 43250 75000 99000 18360 11400 18570 12150 15800 3100 5500 ...
 $ Incidence_Rate_Per_100K : num 89.4 29 25.8 46.8 93.6 90.5 99.1 76.3 88.7 83.5 ...
 $ Mortality_Rate_Per_100K : num 13 5.4 12.7 11.9 14.2 16.5 15.6 10.6 11 12.2 ...
 $ Five_Year_Survival_Pct : num 91 82 66 75.5 87 88 87.5 92.2 91.5 89 ...
 $ Stage_I_II_Pct : int 66 45 30 42 68 72 70 58 70 67 ...
 $ Screening_Program : chr "True" "True" "False" "True" ...
 $ Mammography_Coverage_Pct : num 72.4 22.5 8.2 35 71 55 52 44.7 55 62 ...
 $ Treatment_Access_Score : int 95 72 48 65 93 94 93 90 96 94 ...
```

STEP 2 — Read the data acquisition description :

The data comes from:

- WHO
- GLOBOCAN
- National cancer registries
- WHO GBCI initiative
- CDC, CRUK, ACS

STEP 3 — Identify possible quality problems :

Different countries use different:

- screening systems
- diagnostic accuracy
- reporting standards
- healthcare access
- Population size differences

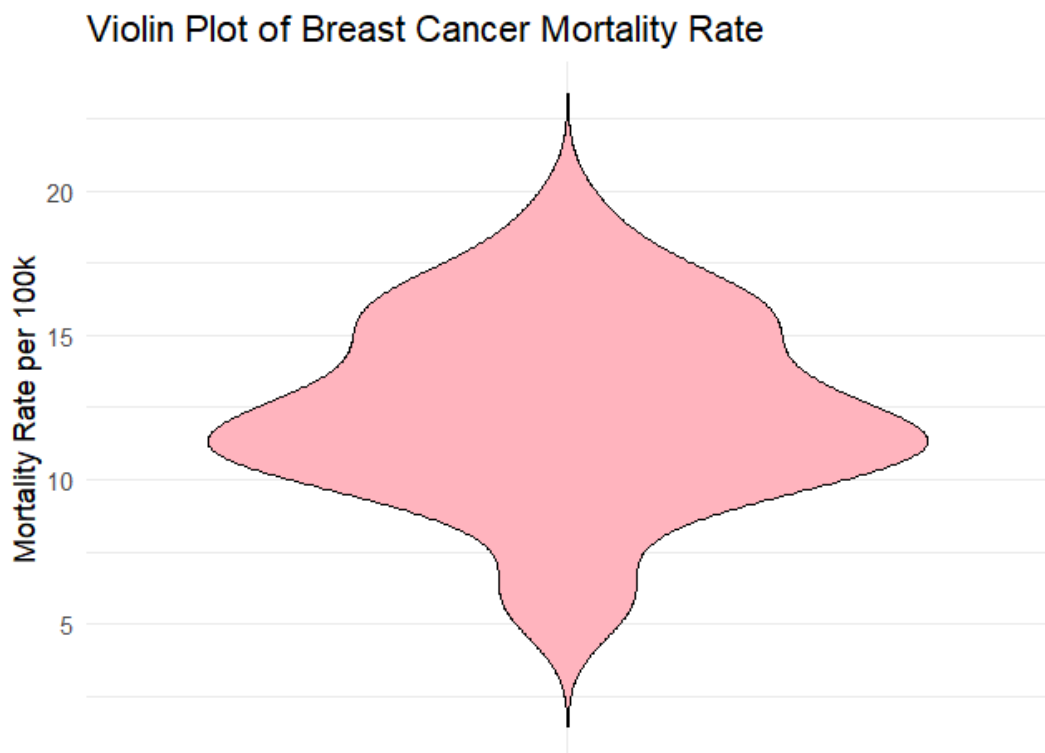
STEP 4 — “Play around with the data :

```
> summary(data)
  Country          Region          Continent      Population_M    New_Cases_2022    Deaths_2022
Length:50      Length:50      Length:50      Min.   :   5.10    Min.   :  1200    Min.   :  250
Class :character Class :character Class :character 1st Qu.:  27.48    1st Qu.:  6400    1st Qu.: 2400
Mode  :character Mode  :character Mode  :character Median :  52.95    Median : 18300    Median : 5350
Mean  : 128.88    Mean  : 39070    Mean  :10704
3rd Qu.: 112.75    3rd Qu.: 32982    3rd Qu.:11300
Max.  :1412.00    Max.  :357000    Max.  :99000

Incidence_Rate_Per_100K Mortality_Rate_Per_100K Five_Year_Survival_Pct Stage_I_II_Pct Screening_Program
Min.   : 16.50          Min.   : 5.40          Min.   :25.00          Min.   :10.00          Length:50
1st Qu.: 31.98          1st Qu.:10.70          1st Qu.:61.25          1st Qu.:30.00          Class :character
Median : 43.20          Median :11.95          Median :77.00          Median :43.50          Mode  :character
Mean   : 55.39          Mean   :12.39          Mean   :71.44          Mean   :44.84
3rd Qu.: 83.42          3rd Qu.:15.22          3rd Qu.:87.50          3rd Qu.:65.75
Max.   :113.20          Max.   :19.40          Max.   :93.60          Max.   :73.00

Mammography_Coverage_Pct Treatment_Access_Score
Min.   : 1.00          Min.   :15.00
1st Qu.:10.50          1st Qu.:48.50
Median :32.50          Median :71.00
Mean   :35.22          Mean   :66.94
3rd Qu.:56.50          3rd Qu.:93.00
Max.   :80.00          Max.   :96.00
```

STEP 5 — Look at distributions:



STEP 6 — Check consistency with background knowledge :

Belgium: Incidence 113.2, mortality 18.4 → high incidence due to screening

Nigeria: Incidence 31.8, mortality 17.0 → low incidence but high mortality due to late detection.

High-income countries → high incidence, lower mortality

Low-income countries → lower incidence, higher mortality

STEP 7 — Identify Outliers, Gaps, Missingness :

- No missing values, but: Screening_Program = False for many countries

#Outliers

Compute IQR-based outliers

```
Q1 <- quantile(data$Mortality_Rate_Per_100K, 0.25)
```

```
Q3 <- quantile(data$Mortality_Rate_Per_100K, 0.75)
```

```
IQR_value <- IQR(data$Mortality_Rate_Per_100K)
```

```
lower_bound <- Q1 - 1.5 * IQR_value
```

```
upper_bound <- Q3 + 1.5 * IQR_value
```

```
outliers <- data[data$Mortality_Rate_Per_100K < lower_bound |  
                 data$Mortality_Rate_Per_100K > upper_bound, ]
```

outliers

```
> outliers  
[1] Country      Region      Continent      Population_M  
[5] New_Cases_2022 Deaths_2022 Incidence_Rate_Per_100K Mortality_Rate_Per_100K  
[9] Five_Year_Survival_Pct Stage_I_II_Pct Screening_Program Mammography_Coverage_Pct  
[13] Treatment_Access_Score  
<0 rows> (or 0-length row.names)  
> colSums(is.na(data))  
      Country      Region      Continent      Population_M  
      0          0          0          0  
New_Cases_2022 Deaths_2022 Incidence_Rate_Per_100K Mortality_Rate_Per_100K  
      0          0          0          0  
Five_Year_Survival_Pct Stage_I_II_Pct Screening_Program Mammography_Coverage_Pct  
      0          0          0          0  
Treatment_Access_Score  
      0
```

STEP 8 — Subgroup analysis

```
> aggregate(Mortality_Rate_Per_100K ~ Continent, data, summary)
Continent Mortality_Rate_Per_100K.Min. Mortality_Rate_Per_100K.1st Qu. Mortality_Rate_Per_100K.Median
1 Africa 10.70000 12.00000 14.00000
2 Americas 8.70000 10.60000 11.50000
3 Asia 5.40000 8.80000 10.60000
4 Europe 10.80000 13.50000 15.60000
5 Oceania 11.00000 11.25000 11.50000
Mortality_Rate_Per_100K.Mean Mortality_Rate_Per_100K.3rd Qu. Mortality_Rate_Per_100K.Max.
1 13.98889 15.70000 17.00000
2 11.72222 12.20000 17.50000
3 10.10588 11.80000 15.60000
4 14.87692 16.00000 19.40000
5 11.50000 11.75000 12.00000
> |
```

8) Missing values in Data:

Missing values in data can be handled by various ways, such as,

1. We can handle it by guessing the values with domain specific knowledge.
2. Imputation by mean, median or mode of the particular feature.
3. We can delete the particular feature if the missing values are more.
4. If the missing values are less, we can ignore the particular row.

Handling missing values according to the data type:

1. **Numeric data:** Use imputation like mean, median, mode.
2. **Categorical data:** We can create a new category as UNKNOWN.
3. **Spatial data:** Domain specific correction is required.